Supplementary Information

Dams mark drought vulnerability rather than buffer it, and storage cannot offset a falling water supply in Chile

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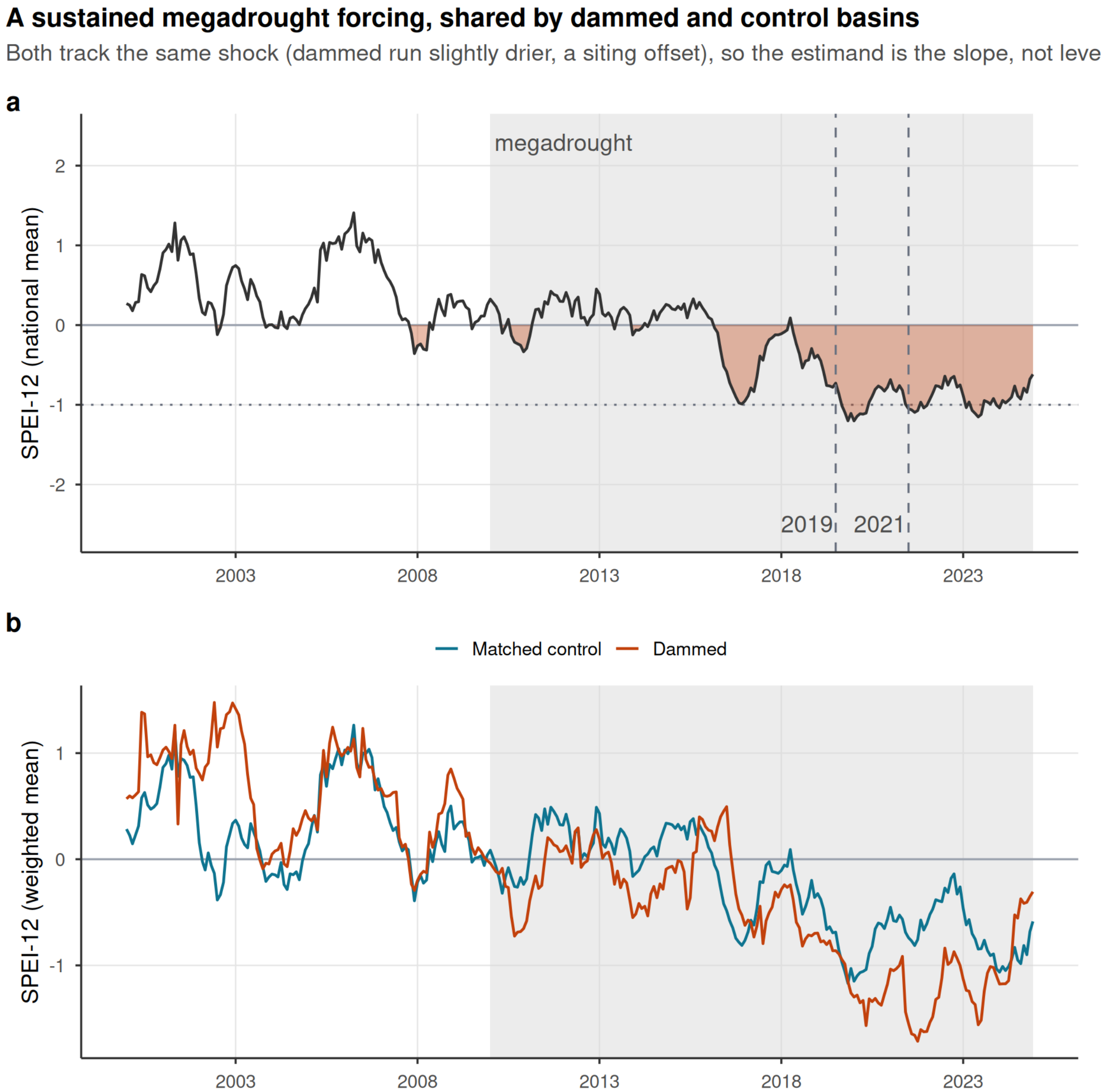
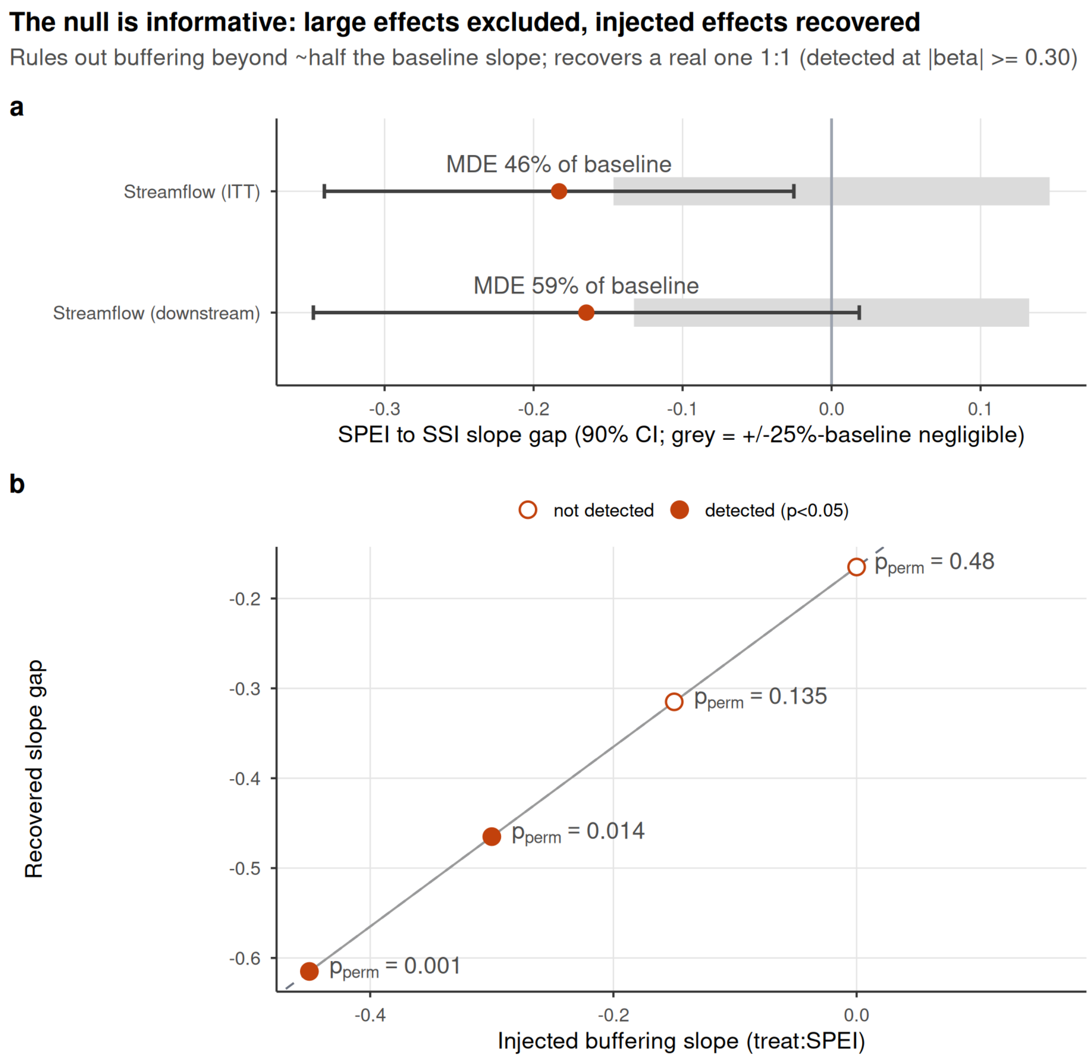
This Supplementary Information collects the supporting design, robustness, and diagnostic results referenced in the main text. Supplementary Figures S1–S3 document the drought forcing, the informativeness of the null, and the siting-confound decomposition; Supplementary Figures S4–S5 give the full study-area map and the matching diagnostics; Supplementary Tables S1–S5 report the equivalence bounds, placebo comparability, positive control, decomposition ladder, and storage-band trends. All items are regenerated by the `targets` pipeline.

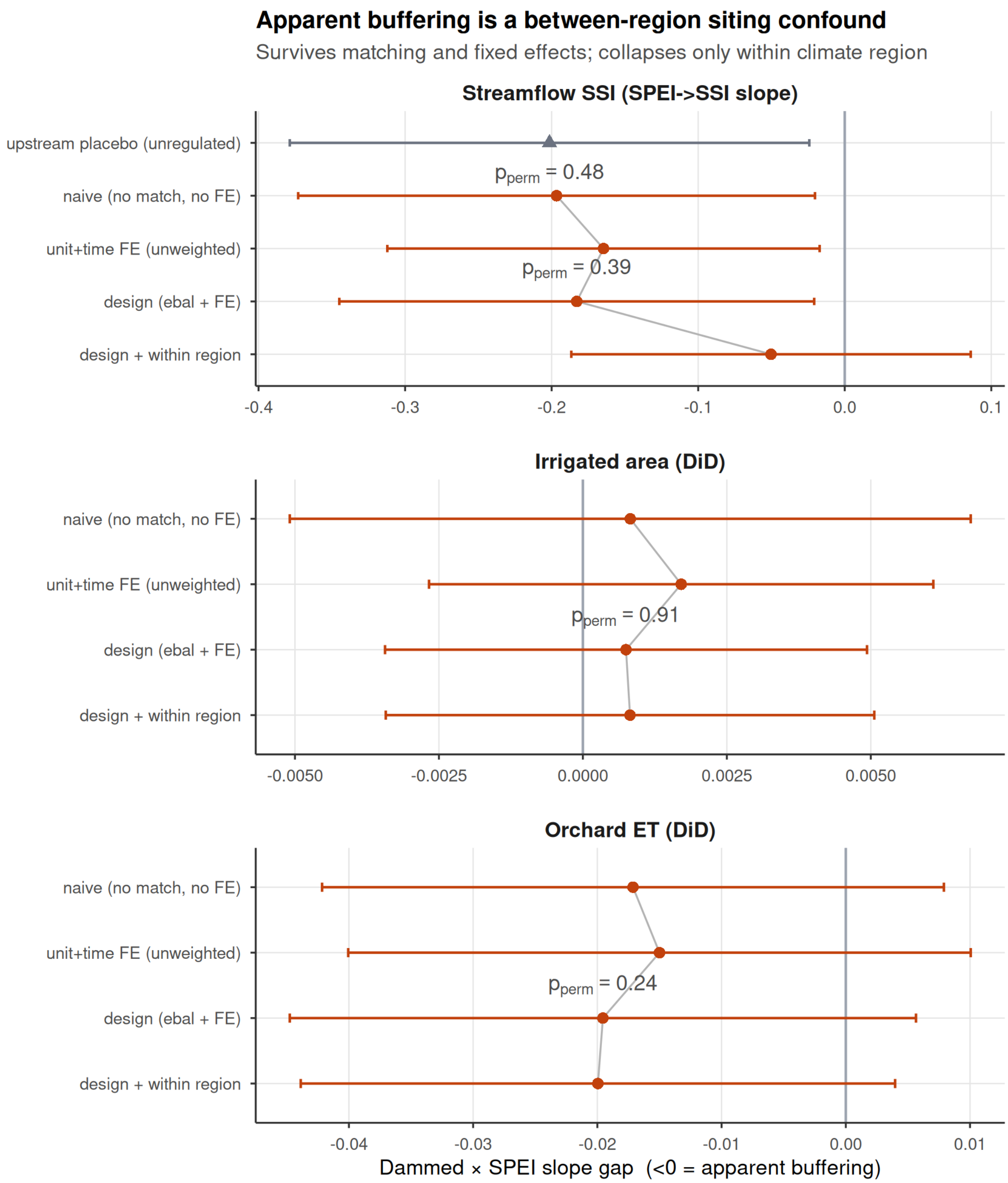
Supplementary Figures



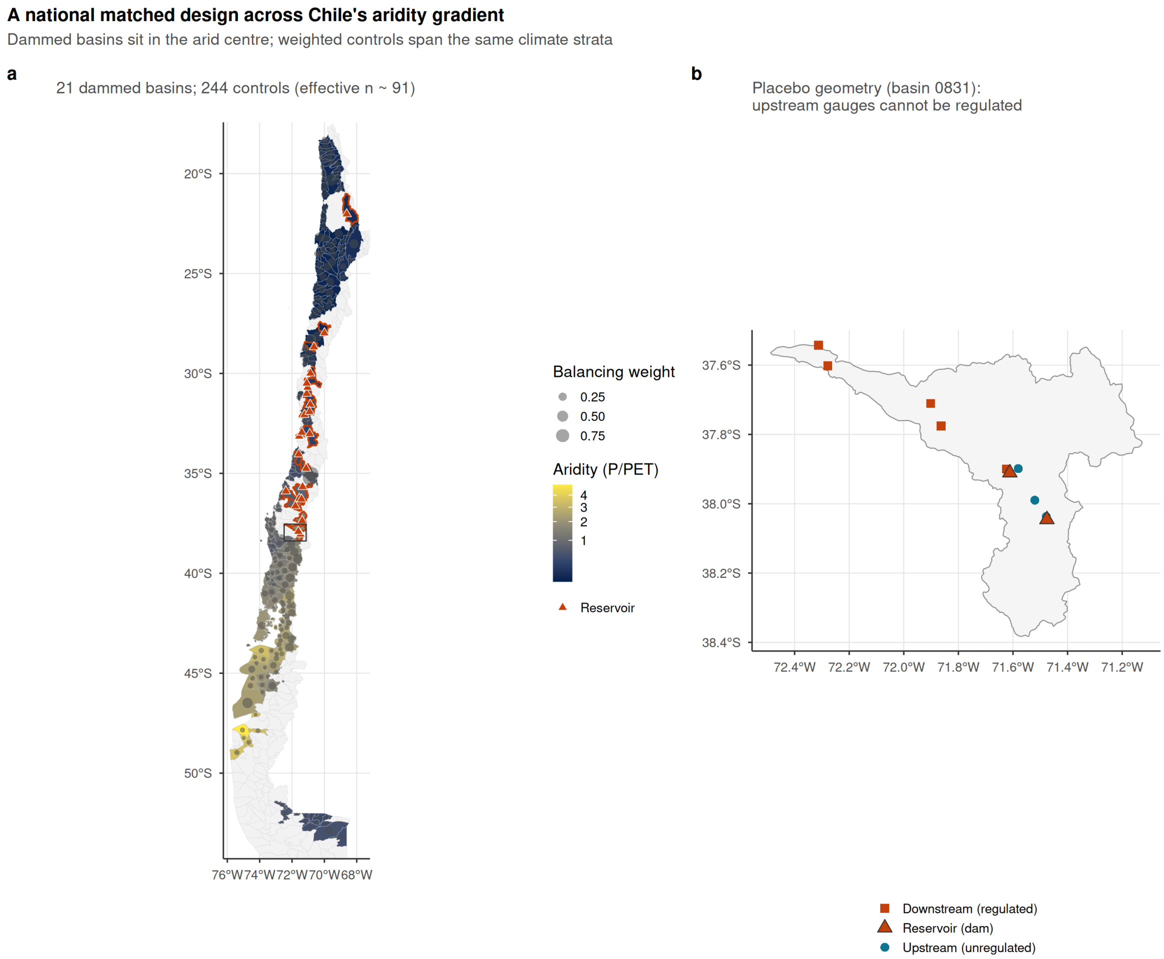
Supplementary Figure S1 | The drought forcing. (a) National mean SPEI-12 across the matched basins, 2000–2024; the negative (drought) excursion is filled and the post-2010 megadrought era shaded, with the 2019 and 2021 hyperdrought winters marked. The deficit is sustained and deep (below −1 for much of 2016–2024), giving the SPEI dose real dynamic range. (b) Entropy-balancing-weighted mean SPEI-12 for dammed versus matched-control basins: both co-move through the same megadrought (a common shock that year fixed effects absorb), with dammed basins running slightly drier, a siting offset that is precisely why the estimand is the differential deficit-to-impact slope rather than a level or calendar-time contrast.



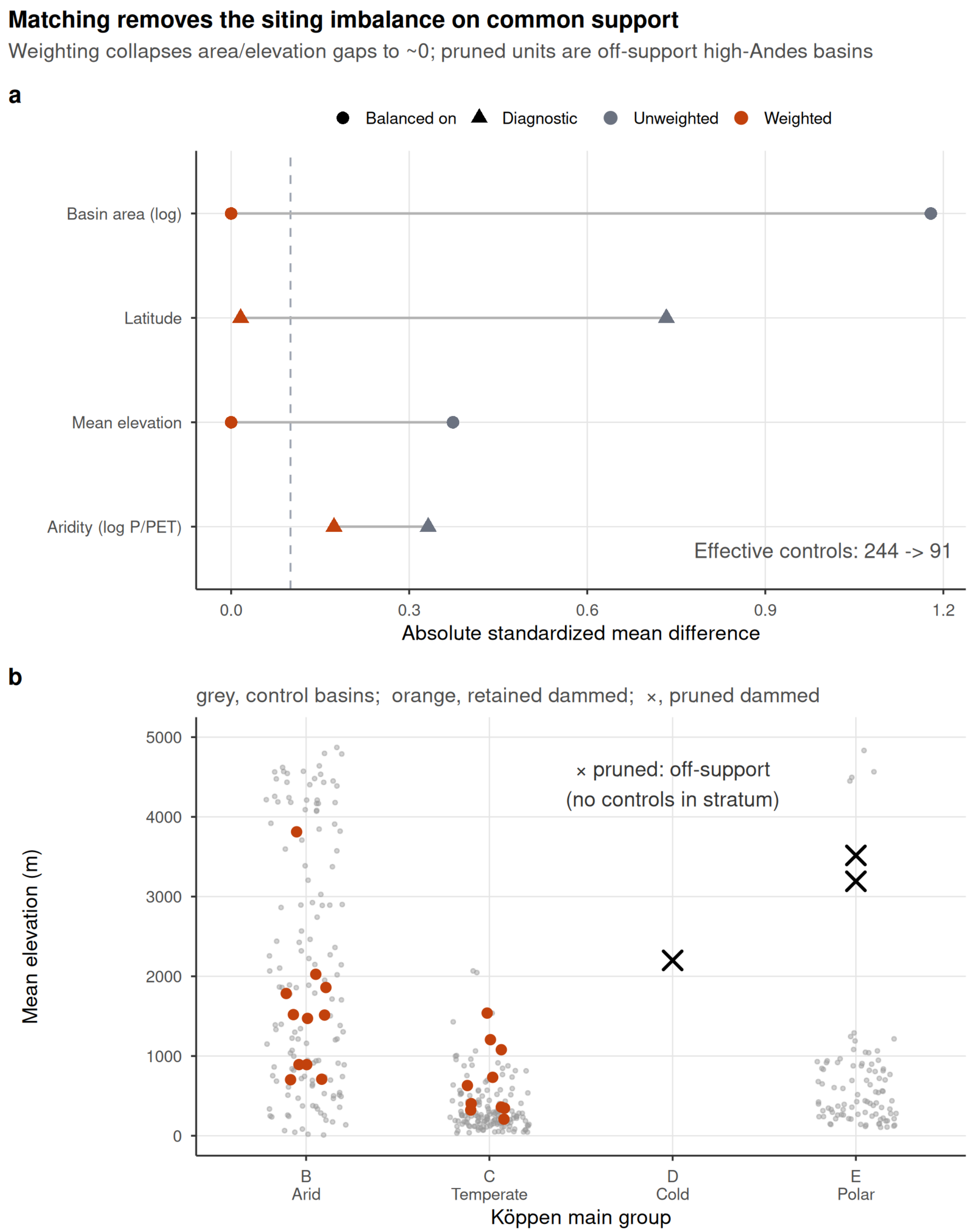
Supplementary Figure S2 | The null is informative, not underpowered blindness. (a) Equivalence / minimum-detectable-effect bounds for the two streamflow outcomes: the observed slope gap (point) and its 90% (TOST) interval against the ±25%-of-baseline negligible region (grey). The 90% intervals exceed the negligible region, so formal equivalence to zero is not claimed, but they exclude attenuations larger than the minimum detectable effect (45% of baseline transmission for the intent-to-treat design, 58% downstream). (b) Positive control: a known buffering slope injected into the treated downstream gauges is recovered one-to-one by the estimator (points on the dashed expected-recovery line), the no-injection case is correctly not flagged, and randomization inference detects the effect (filled points) once the injected slope reaches −0.30. The estimator would therefore reveal a large operational buffering effect if one existed.



Supplementary Figure S3 | Siting-confound decomposition ladder. Dammed-versus-control transmission-slope gap (treat:SPEI; <0 = apparent buffering) estimated by increasingly stringent estimators, for the streamflow SSI slope (top) and the two difference-in-differences proxy outcomes; bars are 95% cluster-robust confidence intervals and the design and within-region rows carry the design's randomization-inference p. Rungs: (1) naive pooled (no matching, no fixed effects, what a conventional dammed-versus-undammed study reports); (2) unit and time fixed effects, unweighted; (3) the design estimator (entropy-balance weights + fixed effects); (4) design with the baseline SPEI-to-outcome slope allowed to vary within climate region. For streamflow the apparent buffering survives rungs (1) to (3) near -0.18 (permutation-null throughout), then collapses toward zero at rung (4), and is equally strong at the unregulated upstream placebo (grey triangle): the signature of a between-region aridity confound, not regulation. The proxy outcomes carry no signal at any rung.



Supplementary Figure S4 | The national matched design across Chile's aridity gradient. (a) Matched sample of subcuencas, each filled by long-term aridity (P/PET, the north-south gradient the design conditions on); dammed (treated) basins are outlined in orange and marked with their reservoir (triangles), and each matched-control basin carries a centroid dot sized by its entropy-balancing weight, so the effective control sample (91) is visible against the nominal 244. Dammed basins concentrate in the arid centre and north; weighted controls span the same climate strata. (b) Within-basin placebo geometry for one representative dammed basin: the reservoir (triangle) with its downstream (regulated) gauges below the dam and upstream (unregulated) gauges above it.



Supplementary Figure S5 | Covariate balance and common support for the matched design. (a) Love plot of the absolute standardized mean difference between dammed and control basins before (unweighted) and after entropy balancing, for the two balanced covariates (log basin area, mean elevation) and two untargeted diagnostics (latitude, log aridity); the dashed line is the conventional 0.1 balance threshold. Weighting collapses the large siting imbalances to ~0, leaving a small residual aridity difference (0.17) that the doubly-robust outcome model absorbs, and reduces the effective control sample from 244 to 91. (b) Climate-stratum common support: mean elevation of control (grey) and dammed (orange, retained) basins within each Köppen main group; the three pruned dammed basins (x) fall in the cold (D) and polar (E) strata, whose controls lie far below them, so they are off-support and excluded rather than matched by extrapolation.

Supplementary Tables

Supplementary Table S1 | Equivalence / minimum-detectable-effect bounds on the headline nulls. Observed slope gap, MDE (smallest effect detectable at 80% power from the permutation-null SD), the 90% (TOST) interval, and whether it lies within the negligible region (±25% of baseline). "baseline" is the untreated (control) SPEI→SSI / SPEI→impact transmission slope from the fixed-effects model. No outcome is formally equivalent to zero; the streamflow design rules out attenuations larger than ~46–58% of baseline transmission.

Outcome	Observed	MDE	90% CI	Equiv. to 0?	p_perm
streamflow SSI (ITT)	-0.183	0.268	[-0.34, -0.0253]	no	0.39
streamflow SSI (downstream)	-0.165	0.312	[-0.348, 0.0186]	no	0.48
irrigated area (DiD)	0.001	0.010	[-0.00506, 0.00656]	no	0.90
orchard ET (DiD)	-0.020	0.046	[-0.0463, 0.00718]	no	0.25

Supplementary Table S2 | Upstream/downstream placebo comparability. Upstream gauges are higher than downstream, and gauge elevation predicts the transmission slope among undammed controls (−0.21 per +1000 m), yet upstream and downstream treated gauges have nearly identical raw transmission slopes, so the equal up/down attenuation is not an elevation artifact.

quantity	value	detail
upstream gauges (elev, mean m)	1003.000	n=28; mean transmission 0.565
downstream gauges (elev, mean m)	459.000	n=40; mean transmission 0.599
control transmission slope vs elevation (per +1000 m)	-0.212	p = 0.00 (n=84 control gauges); ~0 => elevation does not drive transmission

Supplementary Table S3 | Positive control on the streamflow estimator. A known buffering slope (`beta_injected`; negative = buffering) is added to treated downstream units; the estimator recovers it (recovered = baseline −0.17 + beta), the no-effect case is not flagged, and randomization inference detects the effect once it exceeds the minimum detectable effect.

beta_injected	recovered	p_perm	detected
0.00	-0.165	0.480	FALSE
-0.15	-0.315	0.135	FALSE
-0.30	-0.465	0.014	TRUE
-0.45	-0.615	0.001	TRUE

Supplementary Table S4 | Siting-confound decomposition ladder (numeric). The dammed-versus-control transmission-slope gap at each rung of Supplementary Figure S3, for streamflow SSI (ITT) and the two difference-in-differences proxy outcomes, plus the within-basin upstream placebo. `perm_p` is the design's randomization-inference p for the design rung and the placebo.

Outcome	Rung	Slope gap	SE	p_perm
Streamflow SSI (SPEI->SSI slope)	naive (no match, no FE)	-0.197	0.090	-
Streamflow SSI (SPEI->SSI slope)	unit+time FE (unweighted)	-0.165	0.075	-
Streamflow SSI (SPEI->SSI slope)	design (ebal + FE)	-0.183	0.083	0.39
Streamflow SSI (SPEI->SSI slope)	design + within region	-0.050	0.070	-
Irrigated area (DiD)	naive (no match, no FE)	0.001	0.003	-
Irrigated area (DiD)	unit+time FE (unweighted)	0.002	0.002	-
Irrigated area (DiD)	design (ebal + FE)	0.001	0.002	0.91
Irrigated area (DiD)	design + within region	0.001	0.002	-
Orchard ET (DiD)	naive (no match, no FE)	-0.017	0.013	-
Orchard ET (DiD)	unit+time FE (unweighted)	-0.015	0.013	-
Orchard ET (DiD)	design (ebal + FE)	-0.020	0.013	0.24
Orchard ET (DiD)	design + within region	-0.020	0.012	-
Streamflow SSI (SPEI->SSI slope)	upstream placebo (unregulated)	-0.201	0.090	0.48

Supplementary Table S5 | Storage-band trends. Per-year trend slope of the cross-reservoir peak, trough, and peak-trough amplitude (percent of capacity per year; reservoir fixed effects, cluster-robust). Peak and trough both decline while the amplitude trend is not distinguishable from zero, consistent with a supply-side level decline rather than a loss of refill capacity.

Component	Trend (%/yr)	95% CI	p
peak	-1.281	[-2.21, -0.349]	0.013
trough	-1.220	[-1.72, -0.72]	0.000
amplitude	-0.061	[-1.13, 1.01]	0.912